worldskills Asia Taipei2025

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Test Project

[Web Technologies] – Module D (Module 4)

Application – Frontend + Design+ Backend

Taiwan Tourism Module

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Introduction

In this project, we create a web page to promote Taiwan to tourists.
This module has 2 parts: a design implementation part and a backend part.

Overview

A tourism promotion website to showcase Taiwan to international visitors.

This project creates a Taiwan tourism website featuring four regional guides and cultural experiences. Targeting international tourists, it showcases Taiwan's diverse appeal through regional tourism guides for North, Central, South, and East areas and interactive cultural and culinary experience sections, from major attractions like Taipei 101, Jiufen, and Taroko Gorge to authentic cultural and culinary experiences unique to Taiwan.

In this project, you are required to implement the web page design.

There are several sections in the layout:

- Header
- Main Visual
- Regional guide
- Cultural & Culinary Experiences
- Practical Information
- Essential Information
- Contact form
- Footer

Header

Header should be a fixed navigation with the following elements:

- Logo
- Menu (Regional Guide, Experiences, Practical Information, Essential Information, Contact

Main Visual (Hero Section)

The hero section features a full-width background image showcasing a scenic Taiwan landscape. Centered over the image is the main title introducing Taiwan as a travel destination, accompanied by a short tourism slogan that highlights its unique charm and diversity. Below the text, a prominent call-to-action button invites users to “Explore the Regions” and, when clicked, smoothly scrolls the page down to the Regional Guide section.

Regional Guide

The Regional Guide section presents an interactive map of Taiwan, allowing visitors to click on different areas of the island to explore more. Next to the map, four regional cards (all 4 regional cards are always visible) highlight North, Central, South, and East Taiwan. Each card introduces representative attractions and key characteristics of its region—for example, famous cities, natural landscapes, or cultural hotspots. When users hover over a card, additional details appear, providing more in-depth information about must-see spots and unique experiences in that part of Taiwan.

Cultural & Culinary Experiences

The Cultural & Culinary Experiences section lets visitors discover Taiwan's traditions and flavors through two main categories: cultural experiences and culinary experiences. Users can switch between tabs to view tailored content for each category, such as festivals, temples, night markets, and local dishes. A photo gallery in slider format showcases vivid images of these activities and foods, allowing visitors to browse through highlights and get an immersive visual taste of Taiwan's culture and cuisine.

Practical Information

The Practical Information section helps visitors plan their trip with three easy-to-browse tabs: Transportation, Accommodation, and Shopping. Each tab displays clear lists of tips and key details, enhanced with intuitive icons for quick understanding. On mobile devices, the content switches to an accordion format, allowing users to expand and collapse each item for a smooth, space-saving, and responsive browsing experience.

Essential Information

The Essential Information section provides travelers with quick access to official and reliable support for their trip to Taiwan. It highlights the Taiwan Tourism Bureau's contact details, ensuring visitors know exactly where to turn for assistance.

To further support safe and confident travel, the section also features a compact panel with important emergency and safety contacts, including key hotlines and brief guidance for international tourists. Additionally, a downloadable travel guide (pdf format), easy-to-download.

Contact Form

The Contact Form section offers a travel consultation form where visitors can ask questions or request personalized trip advice. The form includes fields for name, email, country, interests, and message, with real-time validation to ensure required information is entered correctly before submission. Behind the scenes, each inquiry is saved together with a

submission timestamp (submitted_at), allowing the system to keep an accurate record of when requests were made.

Footer

The Footer section provides a clear closing area for the website, displaying the site logo alongside essential legal and informational links, including Copyright, Privacy Policy, and Terms and Conditions. On the right, a row of social media icons allows visitors to quickly connect with Taiwan's official tourism channels on platforms like Facebook, Instagram, or others, encouraging further engagement and up-to-date travel inspiration.

Interaction Features

- Smooth scroll: Navigation
- Interactive map: Taiwan map regional clicking
- Tab switching: Experience categories, Practical information
- Image gallery: Experience photo slider display

Usability Features

- Responsive design: Mobile-first (760px boundary)
- Hamburger menu: Mobile navigation

Visual Effects

- Fade-in: Element display on scroll
- Hover effects: Regional cards, Experience cards, Buttons
- Slide animations: Mobile menu

Backend

Dynamic Regional Guide

Transform the static Regional Guide section into a database-driven system

Database Schema

Regions table

- Fields: id, key, name, color, summary
- Records: 4 regions (north, central, south, east)

Attractions table

- Fields: id, region_id (FK → regions.id), name, description
- Records: 3 attractions per region (12 total)

On the server side, the application queries all regions together with their related attractions using a JOIN, then renders the region cards with this dynamic data instead of hard-coded text. The output is structured as semantic HTML that connects logically with the interactive Taiwan map, allowing each region area on the map to correspond to its region card and attractions. This setup makes the Regional Guide easy to maintain, extend, and localize in the future.

Contact Form with Server-Side Processing

Convert the placeholder contact form into a functional server-side form processing system

Database Schema

Submissions table

- Fields: id, name, email, country, interests, message, submitted_at

The Contact Form with Server-Side Processing turns the simple consultation form into a fully functional backend feature that safely records visitor inquiries. A submissions table stores each entry with fields for id, name, email, country, interests, message, and submitted_at (the timestamp of submission). On submission, the server validates all required fields (name, email, and message) and ensures the data is correctly formatted. CSRF protection and input sanitization are implemented to guard against security threats such as cross-site request forgery and malicious input. When the data passes validation, the submission is saved to the database, and the user is presented with a clear success message, confirming that their travel inquiry has been received.

Practical Information Tabs

Transform the Practical Information section from static data into a database-driven tab system

Database Schema

Info_items table

- Fields: id, category, icon, title, body
- Categories: transportation, accommodation, shopping
- Records: 3 items per category (9 total)

On the server side, the application queries the info_items table filtered by category to populate each tab with the relevant tips. These items are then rendered as accordion or <details> elements, providing an accessible and mobile-friendly way to expand and collapse information. The output uses semantic HTML structure, making the content easy to understand for both users and screen readers while keeping the layout flexible and maintainable.

Images Loading Responsiveness

Then web page loads the provided low-resolution image files when the screen is less then 760px width.

Instructions to the Competitor

Please consider accessibility when implementing the design. All images should have proper alternatives. All text content should be readable by screen reader.

This project will be assessed in Google Chrome and accessibility to be assessed by NVDA screen reader from NV Access.

Other Informations

This project will be assessed by using Google Chrome web browser.

You may provide a README file for executing guide if necessary.

Marking Summary

	Sub-Criteria	**Marks**
1	Responsive design & image loading	3.00
2	Section layout implementation	3.00
3	Interaction mechanics	4.00
4	Visual effects	3.00
5	Accessibility & usability	4.00
6	Dynamic Regional Guide	3.00
7	Practical Information tabs	2.00
8	Contact Form backend	2.00
Total		**24.00**